Smart Agri-Doctor

AI-Powered Crop Diagnosis and Weather Alerts for Farmers

Agenda

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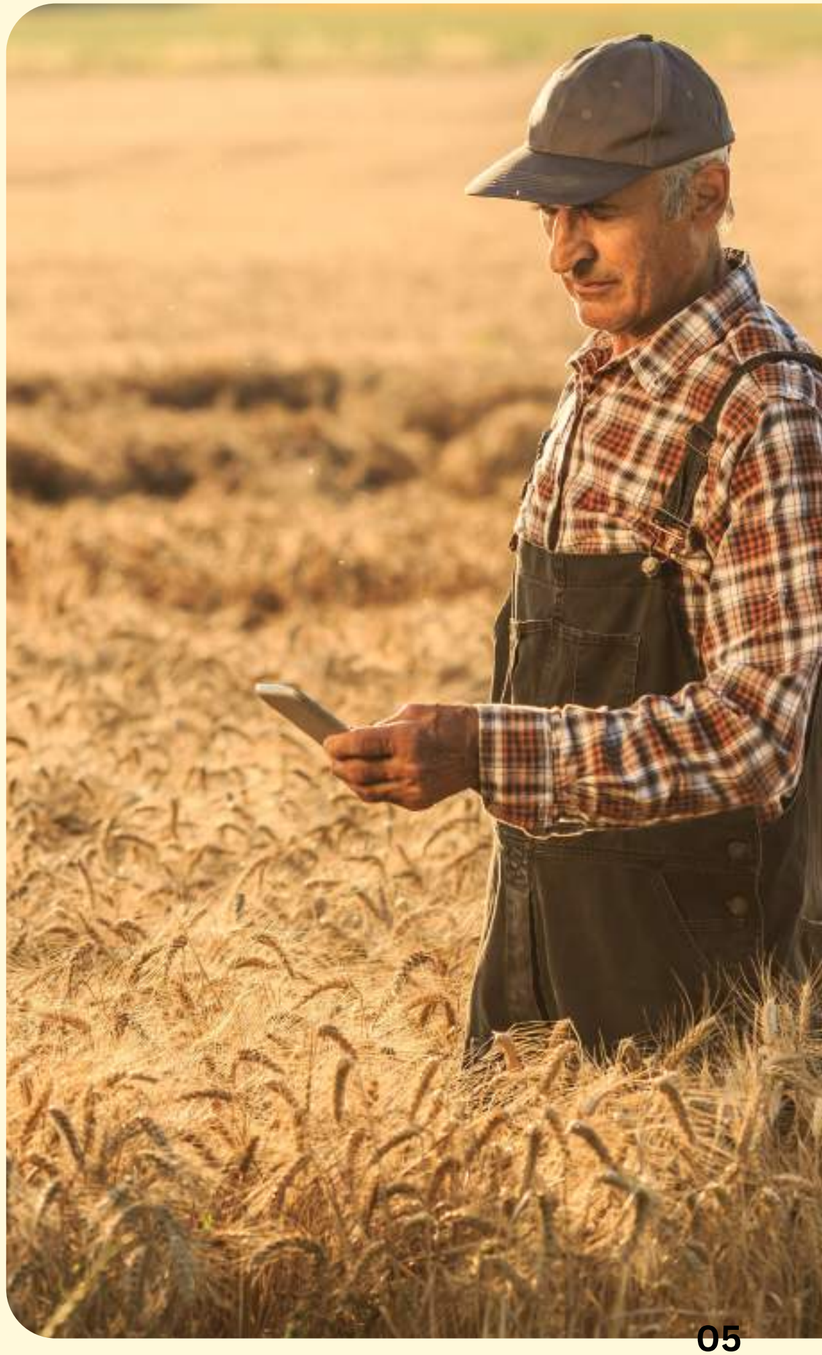
Introduction

Introduction

Smart Agri-Doctor is a mobile application built using Flutter that uses artificial intelligence to help farmers:

- Detect crop diseases by uploading images
- Get treatment suggestions instantly
- Real-time weather update

It is specially designed for local Pakistani farmers, a user-friendly interface that requires no technical background.



Problem Statement

Problem Statement

Pakistani farmers face several critical challenges:

- Lack of access to expert advice in rural areas
- Wrong use of pesticides, leading to crop damage
- Delayed disease detection
- Unpredictable weather affecting crop health

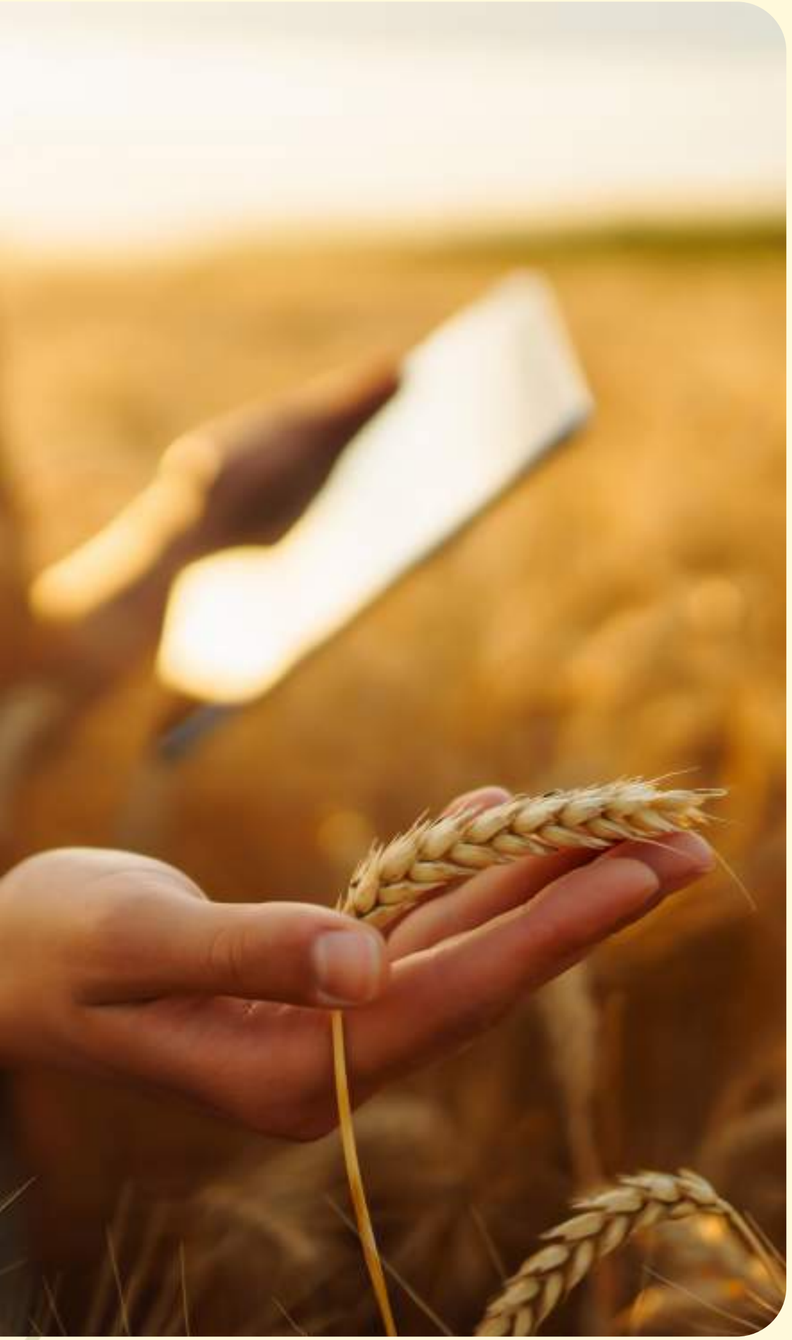
Smart Agri-Doctor solves these by using AI to assist farmers with timely, accurate, and localized guidance.

Objectives

Objectives

- Build a mobile app that diagnoses crop diseases using images
- Use AI models to suggest relevant treatments
- Integrate real-time weather updates using APIs
- Provide localized support (e.g., Urdu) for rural accessibility
- Ensure the app works on low-end Android phones & offline

Procedure



Procedure (App Flow)

- Farmer opens the app
- Selects or takes photo of crop
- App sends image to AI model
- Disease is detected (if any)
- App shows treatment advice
- Weather is fetched using weather API
- Final results are displayed

Core Features

Core Features

- AI-based image analysis for disease detection
- Run locallly on phone without Internet need.
- Real-time weather update.
- Treatment suggestions (fertilizers/pesticides)
- Simple and clean UI for non-technical users.
- Local language.
- Built using Flutter for Android compatibility.

Constraints

Constraints

- Can’t diffrentiate between related and unrelated images.
- Only avalible in Urdu.
- Image quality affects AI accuracy
- Dependent on third-party weather APIs
- Currently limited to pre-selected crops only (Wheat).

Challenges

Challenges

- Finding high-quality, labeled datasets of local crops
- Training accurate AI models with limited resources
- Integrating weather API data for rural areas
- Expensive Weather API’s
- Designing a UI/UX that’s easy for farmers with low literacy
- Ensuring accuracy in real-time crop diagnosis

Tools & Technologies

Tools & Technologies

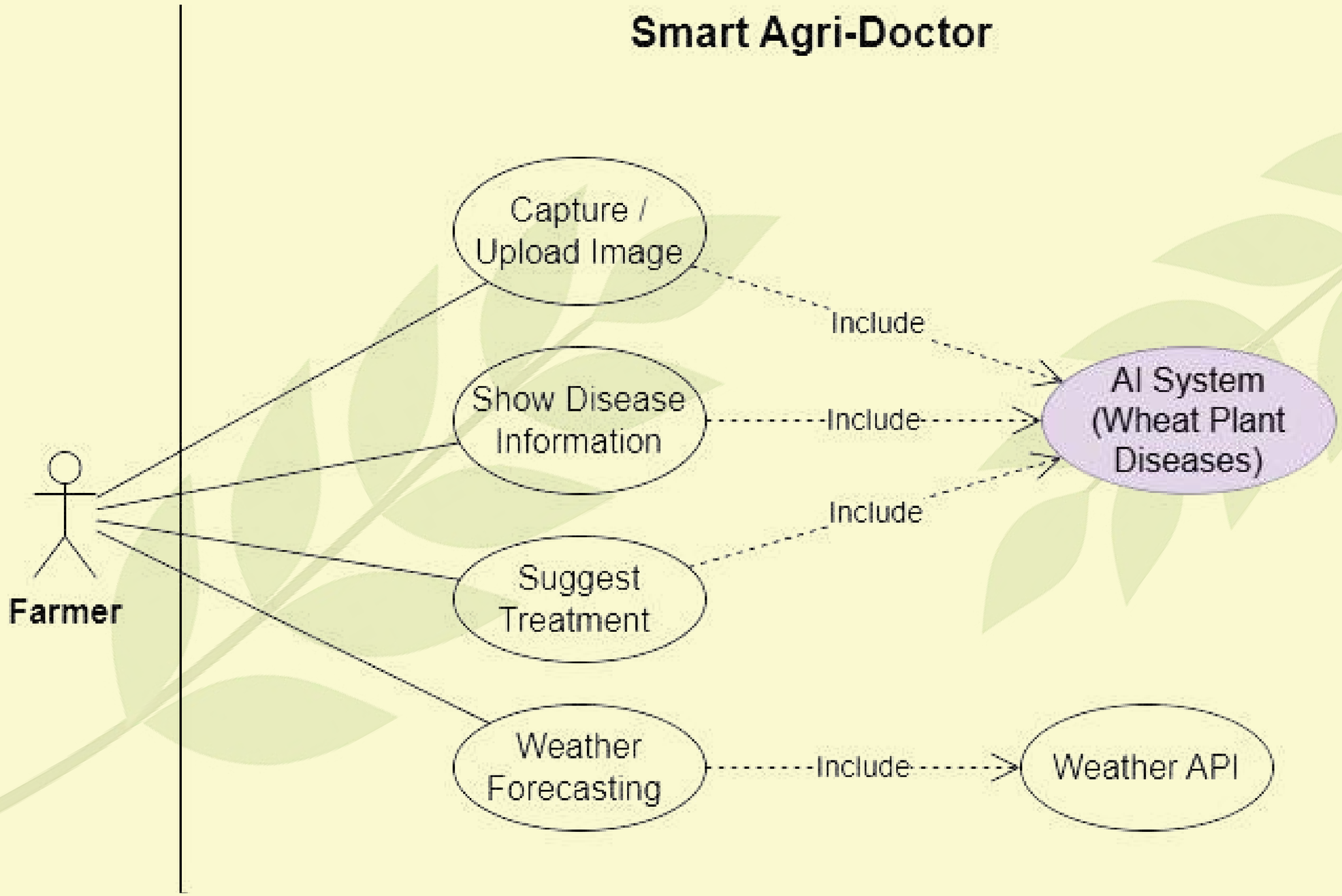
- Frontend: Flutter
- AI Model: Convolutional Neural Network (CNN)
- Dataset: Wheat Plant Diseases Dataset
- Weather API: WeatherApi
- Backend: Custom
- Languages: Dart

Diseases

Diseases

- Aphid [Pest]
- Mite [Pest]
- Stem Fly [Pest]
- Rusts
  - Black Rust / Stem Rust
  - Brown Rust / Leaf Rust
  - Yellow Rust / Stripe Rust
- Smut (Loose, Flag)
- Common Root Rot
- Helminthosporium Leaf Blight / Leaf Blight
- Wheat Blast
- Fusarium head blight / Scab
- Septoria Leaf Blotch
- Spot Blotch
- Tan Spot
- Powdery Mildew

System Diagram


Smart Agri-Doctor
Farmer
Capture / Upload Image
Show Disease Information
Suggest Treatment
Weather Forecasting
Include
Include
Include
Include
AI System (Wheat Plant Diseases)
Weather API

Smart Agri-Doctor
Software Architecture

Processing Layer

Tool Kit
- Image preprocessing
- Crop disease detection model (pre-trained)

Components
- Image Validation
- Disease Detection (AI model)
- Model inference on cloud

Function
- Analyze uploaded image
- Detect crop disease
- Return prediction confidence and disease ID

Input Layer

Platform: Mobile App (Flutter)

Components
- Camera/Image Picker
- Crop Type Selector

Function
- Allow farmer to upload or capture crop images
- Provide crop type selection to improve AI accuracy

Farmer Interface Layer

Platform: Flutter Mobile App

Function
- Upload image
- View diagnostic results
- View treatment suggestions
- View and manage past queries
- Receive weather alerts

Application Logic Layer

Programming Language
- Dart (Flutter)
- Python (for backend AI processing)

Components
- Logic to handle user input
- Diagnosis result manager

Function
- Combine disease + weather info
- Pass processed result to UI layer

Backend & Notification Layer

Platform: Firebase

Components
- Real-time database (save image results, logs)
- Cloud messaging (send notifications)

Function
- Save and retrieve user data and history
- Push treatment suggestions
- Send Weather alert messages to user

Weather API Layer

Tool/Service: OpenWeatherMap API

Function
- Fetch weather data based on user location
- Show temperature, forecast, weather alerts

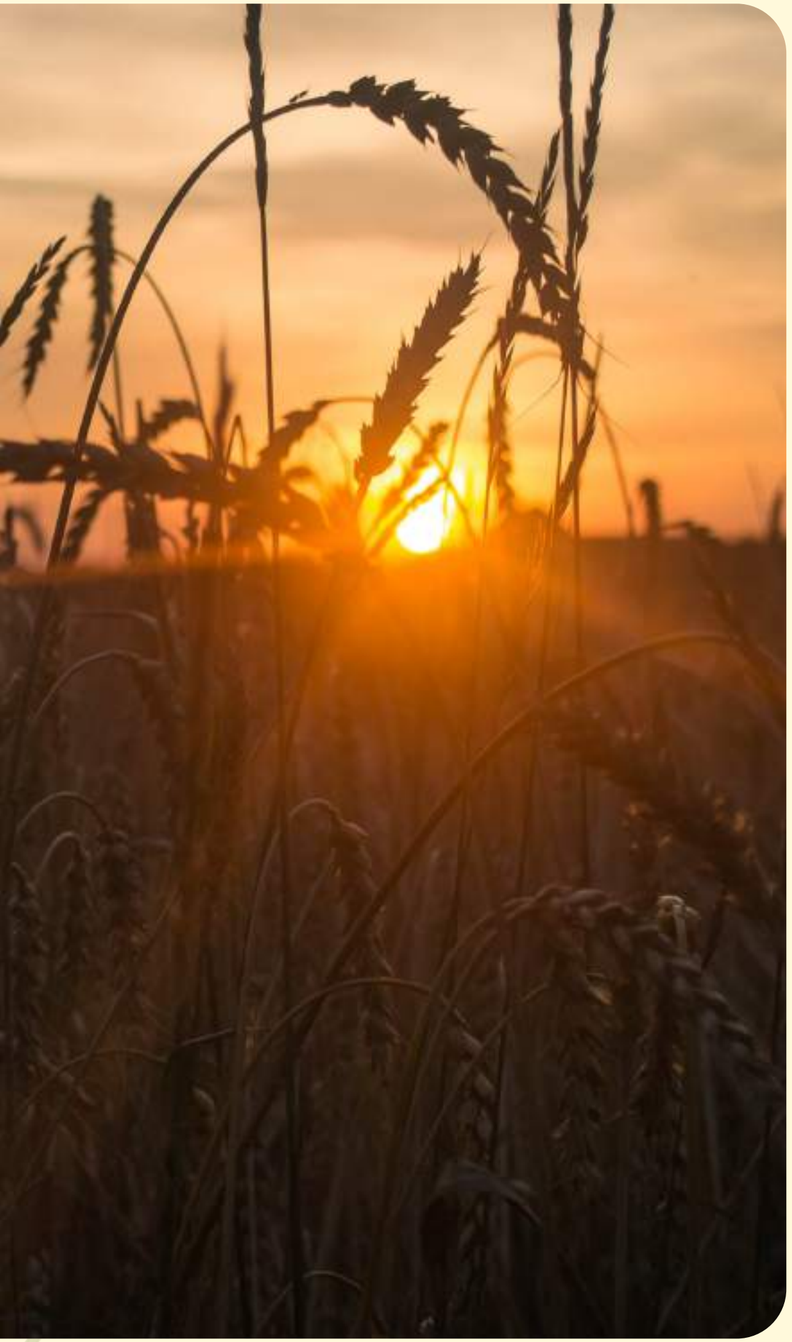
Output / UI Demo

Output & UI/UX

- Upload Image → Diagnosis Result
- Treatment advice screen
- Weather forecast screen
- Final summary screen

Output / UI Demo

Future Work



Future Work

- Expand to more crop types and regions
- Launch iOS version
- Use feedback loop to retrain AI for better accuracy
- Include voice input and voice feedback
- Partner with local agriculture departments for deployment
- Expand Dataset
- Weather Forcast